

Base Rate Neglect Clinic Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	base-rate-neglect-clinic

Big idea: Probability judgements depend on the background rate as well as the new evidence.

Student-Facing Prompt

Inspect Base Rate Neglect Clinic Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Base Rate Neglect Clinic Stimulus A	Student-facing stimulus 1: In 1000 people, 10 have Condition X. The test catches 9 of them but falsely flags 99 people without it. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Base Rate Neglect Clinic Stimulus B	Student-facing stimulus 2: Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Base Rate Neglect Clinic Reveal / Twist	Student-facing stimulus 3: Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Probability judgements depend on the background rate as well as the new evidence.

Actual Lesson Questions

#	Question for students
1	After inspecting Base Rate Neglect Clinic Stimulus A, what is your first knowledge claim?
2	Which exact detail from Base Rate Neglect Clinic Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	How does representation affect mathematical knowledge?

Ready-to-Use Examples

- In 1000 people, 10 have Condition X. The test catches 9 of them but falsely flags 99 people without it.
- Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Present a fictional condition with a low base rate and a fairly accurate test.
2. Ask students to estimate the chance someone has the condition after a positive result.
3. Build the scenario with a 1000-person or 100-person frequency table.
4. Compare intuitive estimates with the frequency result.
5. Debrief why base rates are easy to neglect.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Why can mathematically correct evidence still mislead intuitive knowers?
- When should probability be communicated as percentages and when as natural frequencies?
- How does format affect mathematical understanding?

Knowledge Questions

- How does representation affect mathematical knowledge?
- Why can intuition misread probability?
- What responsibilities come with communicating risk?

Transfer Example

For example, run this activity with a medical test scenario with prevalence, sensitivity, and false positives using $p(h | +)$. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...